



INTERNSPARK

NETWORK PACKET ANALYSIS REPORT

Network traffic analysis identifying TCP handshake behavior, encrypted HTTP/DNS communication patterns, retransmissions, and abnormal local connection failures through packet capture inspection by using Wireshark

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Introduction:

Objective:

The objective of this assessment was to capture and analyze network packets in order to identify communication protocols, TCP connection behavior, and suspicious or unusual network activity.

Scope:

The analysis focused on:

- HTTP/HTTPS traffic
- DNS communication
- TCP handshake behavior
- Packet flags and structures
- Identification of unusual or malformed packets

Tools Used:

Wireshark

Used for:

- Capturing network traffic
- Analyzing packet structures
- Monitoring protocol behavior
- Detecting abnormal traffic patterns

Methodology:

The packet analysis assessment was performed using a structured network monitoring approach. Traffic generated by the local system was captured and analyzed in real time using Wireshark.

The methodology focused on identifying:

- TCP handshake behavior
- HTTP/HTTPS communication
- DNS resolution activity
- Packet structures and flags
- Suspicious or abnormal traffic patterns

Steps involved:

Step 1 — Packet Capture:

Network traffic was captured while performing normal browsing and local application activity. Packet captures included:

- TCP traffic
- HTTPS communication
- DNS queries
- QUIC protocol traffic

The capture process monitored both:

- Localhost communication
- External internet communication

Step 2 — Traffic Filtering:

Wireshark display filters were used to isolate specific packet types for analysis.

TCP Handshake Filter

```
"tcp.flags.syn == 1 && tcp.flags.ack == 0"
```

This filter displayed TCP SYN packets used to initiate network connections.

Step 3 — Protocol Analysis:

Captured packets were analyzed to identify:

- Active protocols
- Packet flow behavior
- Connection establishment sequences
- Encrypted communication sessions

Protocols identified during analysis included:

- TCP
- DNS
- TLSv1.2
- TLSv1.3
- QUIC (HTTP/3)

Step 4 — Packet Structure Inspection:

Individual packets were inspected to analyze:

- Source and destination addresses
- Port numbers
- TCP flags
- TLS handshake messages
- QUIC frame types

Special attention was given to:

- SYN packets
- ACK responses
- Reset packets (RST)
- Retransmissions
- Encrypted application data

Step 5 — Identification of Abnormal Traffic:

Captured traffic was examined for unusual behavior such as:

- Failed TCP handshakes
- Continuous retransmissions
- Port reuse warnings
- Reset connections
- Packet capture anomalies

Observed warnings included:

[TCP Retransmission]

[TCP Spurious Retransmission]

[TCP Port numbers reused]

[TCP Previous segment not captured]

Step 6 — Risk Evaluation:

The identified behaviors were analyzed to determine:

- Whether traffic was normal or suspicious
- Possible causes of failures
- Potential network or application issues
- Security impact and risk level

Traffic behavior was categorized as:

- Normal encrypted communication
- Local service failure
- Network instability
- Capture-related anomalies

TCP Handshake Analysis:

Capture Filter Used:

“tcp.flags.syn == 1 && tcp.flags.ack == 0”

Observations:

The packet capture identified multiple TCP SYN packets originating from:

- 127.0.0.1
- ::1

Destination ports observed:

- 1521 (Oracle Database)
- 8080 (Web Server)
- 3000 (Development Server)

Behaviors Identified:

- Some SYN packets received no response
- Some connections received immediate: RST, ACK

responses indicating failed connections.

HTTP / HTTPS Traffic Analysis:

Ports Observed:

- 443 (HTTPS)

Protocols Identified:

- TLSv1.2
- TLSv1.3
- QUIC (HTTP/3)

TLS Handshake Observations:

Successful encrypted communication included:

- Client Hello
- Server Hello
- Encrypted Application Data

Server Name Indication (SNI) Detected:

- www.google.com
- fonts.gstatic.com
- www.gstatic.com

This confirms successful secure communication with external servers.

DNS Traffic Analysis:

DNS queries were observed before external encrypted communication.

Domain Resolution Activity:

DNS lookups were identified for:

- Google-related services
- External HTTPS resources

These resolutions were followed by TLS and QUIC communication sessions.

Packet Structure and Flag Analysis:

TCP Flags Observed:

Flag	Meaning	Observed Behavior
SYN	Start connection	Frequently observed
ACK	Acknowledgment	Present in responses
RST	Reset connection	Seen in failed local connections
PSH	Push data	Seen during HTTPS traffic
FIN	Close connection	Seen during retransmissions

Packet Behavior Analysis:

- Normal TCP Handshake:
Successful external communication followed the standard TCP handshake process:

SYN → SYN-ACK → ACK

This indicates healthy network communication.

- Failed TCP Handshake:
Some local services displayed failed connection attempts:

SYN → RST, ACK

This indicates that no active service was listening on the destination port.

- TCP Retransmissions:
The following retransmission events were observed:
 - [TCP Retransmission]
 - [TCP Spurious Retransmission]These indicate repeated packet transmission attempts.

QUIC Packet Structure:

QUIC traffic was identified as UDP-based encrypted communication.

QUIC Frame Types Observed:

- CRYPTO
- PING
- PADDING

TLS Packet Structure:

TLS traffic included:

- Client Hello
- Server Hello
- Change Cipher Spec
- Encrypted Application Data

This confirms successful encrypted sessions.

Unusual or Malformed Packet Analysis:

Issue 1: Broken TCP Handshakes:

Observation:
Repeated SYN packets were observed without successful handshake completion.
Immediate responses included: RST, ACK

Analysis:
This behavior indicates:

- Closed ports
- Stopped services
- No active listeners on:
- Port 8080
- Port 1521

Risk Level:
Medium

Issue 2: TCP Port Reuse:

Observation:
Wireshark identified:
[TCP Port numbers reused]

Analysis:
This indicates:

- Rapid retry attempts
- Repeated failed connections
- Possible application retry loops

Risk Level:
Medium

Issue 3: TCP Retransmissions:

Observation:
Normal and spurious retransmissions were detected.

Analysis:
Possible causes include:

- Packet loss
- Temporary network delay
- Wi-Fi instability
- Congestion

Risk Level:
Low

Additional Warnings Observed during analysis:

Warnings:

[TCP Previous segment not captured]
[Ignored Unknown Record]

Analysis:
These warnings are generally related to:

- Packet capture limitations
- Missing captured packets
- Non-malicious traffic anomalies

No malicious activity was identified.

Executive Summary:

The packet capture analysis identified two major network behaviors:

1. Failed local TCP connections to ports 8080 and 1521
2. Successful secure external communication using TLS and QUIC protocols

Technical Findings:

Local Service Failures:
Continuous SYN packets followed by reset responses indicate:

- Services not running
- Misconfigured applications
- Closed listening ports

Application Retry Behavior:
Rapid retries and TCP port reuse suggest:

- Poor retry handling
- Possible infinite connection retry loops

Healthy External Traffic:
Successful TLS and QUIC handshakes confirm:

- Proper encrypted communication
- Secure internet connectivity

Network Stability:
Minor retransmissions indicate:

- Temporary packet loss
- Normal network delays

No evidence of malicious traffic was observed.

Risk Classification:

Issue	Risk Level
Local service failure	Medium
TCP retransmissions	Low
External encrypted communication	No Risk

Recommendations:

Service Verification:

Start and verify required services:

- Oracle Database (1521)
- Web Server (8080)

Port Configuration:

Verify:

- Listening ports
- Service bindings
- Firewall rules

Application Improvements:

- Fix retry logic
- Prevent continuous reconnection attempts
- Implement proper timeout handling

Network Monitoring:

- Monitor retransmission frequency
- Check network stability if retransmissions increase

Conclusion:

The packet analysis revealed failed local service connections caused by unavailable or closed ports, along with repeated retry attempts generated by local applications. Despite local service failures, external internet communication was functioning securely using TLS and QUIC protocols.

Overall, the environment demonstrated:

- Secure encrypted external communication
- Minor network retransmissions
- Local service configuration issues requiring remediation

